Derivation appendix: 05 — S_3-3HDM scalar parameter space in the soft-broken vacuum

the soft-broken vacuum: tadpoles, fundamental domain, spectrum

1. The soft-broken tadpoles fix $(\mu_0^2, \mu_1^2, m_{D1}^2)$ for a general vacuum

derived here — §2

Independent checks

- solving for m_{D2}^2 instead divides by $v_1^2 - v_2^2$ (singular at $\varphi = 45°$); m_{D1}^2 divides by $v_1 v_2$ only
- at $v_2 = \sqrt{3} v_1$ with the free soft terms off, $m_{D1}^2 = 0$

Would be falsified by: a non-zero m_{D1}^2 on the aligned, zero-soft vacuum

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2. For a general vacuum R still isolates both Goldstones; the CP-even sector is a full 3×3 with no gauge-phobic state

derived here — §4

Independent checks

- first rows of $R^{\mathsf{T}} M_{A,C} R$ vanish exactly at a rational point
- masses match mpmath eigsy of the un-rotated matrices, 60 points
- $\sum_k g_{h_k VV}^2 / g_{\mathrm{SM}}^2 = 1$ to 10^{-12}

Would be falsified by: a non-zero Goldstone row, or a mass disagreeing with brute force

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3. $\varphi \in (0, \pi/3)$ is a fundamental domain: $\varphi \to 2\pi/3 - \varphi$ with P on both spurion doublets is the same physics

derived here — §3

Independent checks

- identical spectra and solved m_{D1}^2 at 200 random image pairs
- leaving the **2** spurion untransformed breaks it

Would be falsified by: a spectrum differing between a point and its image

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4. The draft-basis vacuum ratio is $r = -\cot(\varphi - \pi/6)$

derived here — §3

Independent checks

- agrees with fermions.tex_basis_map at 40 angles
- $|r(\pi/3)| = \sqrt{3}$; the quark-fit $|r|$ lies inside the domain

Would be falsified by: disagreement with the verified basis map

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5. Soft terms need not be small: they renormalize only in proportion to themselves

imported — §1

Independent checks

- technical naturalness ['tHooft80]: small soft terms are stable, large ones equally consistent

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